The type of extensional well-founded relations is a set

The Univalent Foundations Program

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Editorial note. The counted statement is the unlabelled author thm environment at source lines 1072-1074 and its proof environment opens on line 1075, the next line; the pairing is by adjacency and the retained proof range 1075-1087 is the whole environment.

Editorial note. The environment carries no author label and the pinned book label-number dump has no entry for it, so no counter value is imposed and the counted theorem prints with the wrapper numbering; the original number is not claimed.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command at all.

Editorial note. The proof contains an index entry inside its range (automorphism of extensional well-founded relations); it is retained verbatim and produces no visible output.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the type of extensional well-founded relations is a set (unlabelled author theorem opening at line 1072).

Allowed standard prerequisites (editorial).

- Equality of structures through univalence. By univalence an equality of extensional well-founded relations reduces to an equivalence of the underlying structures, so the type is a set as soon as every automorphism is the identity.. It is the reduction performed at the start of the proof.
- Extensionality of the relation. Two elements of an extensional well-founded relation are equal as soon as they have the same predecessors.. It turns the goal for one element into a pointwise comparison of the predecessor conditions.
- Induction on an accessible element. To prove a family over a well-founded relation it suffices to prove it for an element assuming it for all strictly smaller elements.. It is the induction on the relation that the proof performs.

Theorem 0.0.1. *The type of extensional well-founded relations is a set.*

Proof. By the univalence axiom, it suffices to show that if $(A, <)$ is extensional and well-founded and $f : (A, <) \cong (A, <)$, then $f = \mathsf{id}_A$. We prove by induction on $<$ that $f(a) = a$ for all $a : A$. The inductive hypothesis is that for all $a' < a$, we have $f(a') = a'$.

Now since A is extensional, to conclude $f(a) = a$ it is sufficient to show

$$\forall(c : A).\, (c < f(a)) \Leftrightarrow (c < a).$$

However, since f is an automorphism, we have $(c < a) \Leftrightarrow (f(c) < f(a))$. But $c < a$ implies $f(c) = c$ by the inductive hypothesis, so $(c < a) \to (c < f(a))$. On the other hand, if $c < f(a)$, then $f^{-1}(c) < a$, and so $c = f(f^{-1}(c)) = f^{-1}(c)$ by the inductive hypothesis again; thus $c < a$. Therefore, we have $(c < a) \Leftrightarrow (c < f(a))$ for any $c : A$, so $f(a) = a$. □